



Figure 1: Areas affected by cloud monitoring

system logs); supports remote monitoring; enables auto-discovery of system resources, identifies problems, performs root cause analysis, and offers security through access control.

Ganglia

Ganglia is a monitoring tool for high performance computing systems such as private clouds, public clouds, clusters, and grids. The Ganglia system contains: 1) two unique daemons, 2) a PHP-based Web front-end, and 3) other small programs. Gmond, a multi threaded daemon, runs on each node to monitor changes in the host state, announce applicable changes, listen to the state of all Ganglia nodes via a unicast or multicast channel based on installation, and respond to requests. At regular intervals, Ganglia Meta Daemon polls a collection of data sources, parses the XML, saves all metrics to round-robin databases and exports the aggregated XML. The Ganglia Web front-end is written in PHP and uses graphs generated by gmetad, and provides the collected information like CPU utilisation for the past day, week, month, or year.

Multicast mode is the default setting in Ganglia installation and is the simplest to set up, providing redundancy. Public Cloud Environments such as Amazon's AWS EC2 do not support multicast, so unicast mode installation is the only set-up option available. Eucalyptus is an open source product for building AWS compatible private clouds; its open source version does not provide built-in monitoring but that can be achieved with Ganglia.

The Eucalyptus source package includes scripts that can be used with third party tools such as Ganglia to enable Eucalyptus-specific monitoring on a pre-defined number of hosts.

Nagios

Open source network monitoring and infrastructure monitoring tool Nagios provides monitoring and reporting for network services and host resources. Nagios Core is an

open source, infrastructure monitoring system that enables organisations to diagnose IT infrastructure problems before they have an adverse effect on critical business processes.

Nagios provides monitoring of Cloud Resources, such as compute, storage and network services. Nagios is capable of monitoring virtual servers and OSs in both the physical and virtual environment. With Nagios, it is easy to identify issues in the cloud environment, detect network outages and check application availability.

The Nagios cloud monitoring tool offers multiple benefits such as high availability, fault tolerance and data availability. Nagios provides monitoring of public cloud services such as Amazon EC2 (Elastic Compute Cloud), Amazon S3 (Simple Storage Service), etc. Eucalyptus is a product for building private and hybrid clouds; its open source version does not provide built-in monitoring but that can be achieved with Nagios, using scripts in the Extras directory, along with third party tools, for example, to interact with Nagios. Nagios allows you to write plug-ins in just about any language and run them on remote servers.

OpenNMS

OpenNMS is a network management platform which also has an open source model. It targets organisations that need scalable network management. OpenNMS supports SNMP natively, as well as common service checks. It is a new enterprise grade monitoring system. It employs a console that allows the OpenNMS daemon to communicate network status updates with the front-end engine in real time.

Zabbix

Zabbix is an open source network monitoring tool that can be used to automatically collect and parse data from monitored cloud resources so that administrators can verify the availability and see trends in network performance. It also provides distributed monitoring with centralised Web administration, a high level of performance and capacity, JMX monitoring, SLAs and ITIL KPI metrics on reporting, as well as agent-less monitoring. 

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